

NAME

chmod, fchmod, fchmodat – change permissions of a file

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sys/stat.h>
```

```
int chmod(const char *pathname, mode_t mode);
```

```
int fchmod(int fd, mode_t mode);
```

```
#include <fcntl.h>      /* Definition of AT_* constants */
```

```
#include <sys/stat.h>
```

```
int fchmodat(int dirfd, const char *pathname, mode_t mode, int flags);
```

Feature Test Macro Requirements for glibc (see **feature_test_macros(7)**):

fchmod():

Since glibc 2.24:

```
_POSIX_C_SOURCE >= 199309L
```

glibc 2.19 to glibc 2.23

```
_POSIX_C_SOURCE
```

glibc 2.16 to glibc 2.19:

```
_BSD_SOURCE || _POSIX_C_SOURCE
```

glibc 2.12 to glibc 2.16:

```
_BSD_SOURCE || _XOPEN_SOURCE >= 500
```

```
|| _POSIX_C_SOURCE >= 200809L
```

glibc 2.11 and earlier:

```
_BSD_SOURCE || _XOPEN_SOURCE >= 500
```

fchmodat():

Since glibc 2.10:

```
_POSIX_C_SOURCE >= 200809L
```

Before glibc 2.10:

```
_ATFILE_SOURCE
```

DESCRIPTION

The **chmod()** and **fchmod()** system calls change a file's mode bits. (The file mode consists of the file permission bits plus the set-user-ID, set-group-ID, and sticky bits.) These system calls differ only in how the file is specified:

- **chmod()** changes the mode of the file specified whose pathname is given in *pathname*, which is dereferenced if it is a symbolic link.
- **fchmod()** changes the mode of the file referred to by the open file descriptor *fd*.

The new file mode is specified in *mode*, which is a bit mask created by ORing together zero or more of the following:

S_ISUID (04000) set-user-ID (set process effective user ID on **execve(2)**)

S_ISGID (02000) set-group-ID (set process effective group ID on **execve(2)**; mandatory locking, as described in **fcntl(2)**; take a new file's group from parent directory, as described in **chown(2)** and **mkdir(2)**)

S_ISVTX (01000) sticky bit (restricted deletion flag, as described in **unlink(2)**)

S_IRUSR (00400) read by owner

S_IWUSR (00200) write by owner

S_IXUSR (00100) execute/search by owner ("search" applies for directories, and means that entries within the directory can be accessed)

S_IRGRP (00040) read by group

S_IWGRP (00020) write by group

S_IXGRP (00010) execute/search by group

S_IROTH (00004) read by others

S_IWOTH (00002) write by others

S_IXOTH (00001) execute/search by others

The effective UID of the calling process must match the owner of the file, or the process must be privileged (Linux: it must have the **CAP_FOWNER** capability).

If the calling process is not privileged (Linux: does not have the **CAP_FSETID** capability), and the group of the file does not match the effective group ID of the process or one of its supplementary group IDs, the **S_ISGID** bit will be turned off, but this will not cause an error to be returned.

As a security measure, depending on the filesystem, the set-user-ID and set-group-ID execution bits may be turned off if a file is written. (On Linux, this occurs if the writing process does not have the **CAP_FSETID** capability.) On some filesystems, only the superuser can set the sticky bit, which may have a special meaning. For the sticky bit, and for set-user-ID and set-group-ID bits on directories, see **inode(7)**.

On NFS filesystems, restricting the permissions will immediately influence already open files, because the access control is done on the server, but open files are maintained by the client. Widening the permissions may be delayed for other clients if attribute caching is enabled on them.

fchmodat()

The **fchmodat()** system call operates in exactly the same way as **chmod()**, except for the differences described here.

If the *pathname* given in *pathname* is relative, then it is interpreted relative to the directory referred to by the file descriptor *dirfd* (rather than relative to the current working directory of the calling process, as is done by **chmod()** for a relative *pathname*).

If *pathname* is relative and *dirfd* is the special value **AT_FDCWD**, then *pathname* is interpreted relative to the current working directory of the calling process (like **chmod()**).

If *pathname* is absolute, then *dirfd* is ignored.

flags can either be 0, or include the following flag:

AT_SYMLINK_NOFOLLOW

If *pathname* is a symbolic link, do not dereference it: instead operate on the link itself. This flag is not currently implemented.

See **openat(2)** for an explanation of the need for **fchmodat()**.

RETURN VALUE

On success, zero is returned. On error, -1 is returned, and *errno* is set to indicate the error.

ERRORS

Depending on the filesystem, errors other than those listed below can be returned.

The more general errors for **chmod()** are listed below:

EACCES

Search permission is denied on a component of the path prefix. (See also **path_resolution(7)**.)

EBADF

(**fchmod()**) The file descriptor *fd* is not valid.

EBADF

(**fchmodat**()) *pathname* is relative but *dirfd* is neither **AT_FDCWD** nor a valid file descriptor.

EFAULT

pathname points outside your accessible address space.

EINVAL

(**fchmodat**()) Invalid flag specified in *flags*.

EIO

An I/O error occurred.

ELOOP

Too many symbolic links were encountered in resolving *pathname*.

ENAMETOOLONG

pathname is too long.

ENOENT

The file does not exist.

ENOMEM

Insufficient kernel memory was available.

ENOTDIR

A component of the path prefix is not a directory.

ENOTDIR

(**fchmodat**()) *pathname* is relative and *dirfd* is a file descriptor referring to a file other than a directory.

ENOTSUP

(**fchmodat**()) *flags* specified **AT_SYMLINK_NOFOLLOW**, which is not supported.

EPERM

The effective UID does not match the owner of the file, and the process is not privileged (Linux: it does not have the **CAP_FOWNER** capability).

EPERM

The file is marked immutable or append-only. (See **ioctl_iflags(2)**.)

EROFS

The named file resides on a read-only filesystem.

VERSIONS**C library/kernel differences**

The GNU C library **fchmodat**() wrapper function implements the POSIX-specified interface described in this page. This interface differs from the underlying Linux system call, which does *not* have a *flags* argument.

glibc notes

On older kernels where **fchmodat**() is unavailable, the glibc wrapper function falls back to the use of **chmod**(). When *pathname* is a relative pathname, glibc constructs a pathname based on the symbolic link in */proc/self/fd* that corresponds to the *dirfd* argument.

STANDARDS

POSIX.1-2008.

HISTORY**chmod()****fchmod()**

4.4BSD, SVr4, POSIX.1-2001.

fchmodat()

POSIX.1-2008. Linux 2.6.16, glibc 2.4.

SEE ALSO**chmod(1), chown(2), execve(2), open(2), stat(2), inode(7), path_resolution(7), symlink(7)**